Inventory Management Practices in Private Healthcare Facilities in Nairobi County

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Abstract

Universal health coverage (UHC) is an integral part of the United Nations sustainable development goals. The private sector plays a prominent role in achieving UHC, being the primary source of essential medicines for many people. However, many private healthcare facilities in low- and middle-income countries (LMICs) have insufficient stocks of essential medicines. At the same time, we find that these same facilities carry excessive quantities of certain drugs, leading to losses due to obsolescence. This suggests poor inventory control. To propose potential remedies, it is key to fully understand the underlying causes. In semi-structured interviews with facility managers from private healthcare facilities in Nairobi we asked them about their 1) inventory control systems, 2) inventory control skills, 3) time/human resource constraints, 4) budget constraints, 5) motivations for inventory control, and 6) suppliers. Our results show that the problems are strongly driven by resource limitations (budget and time/human resources), managerial issues (relating to skills and systems), and market mechanisms that limit overage and underage costs. Unavailability at the supplier level, and motivations for inventory control, are only minor issues. We argue that these key causes are strongly interlinked and stem from deeper issues in the market and regulatory environment. Our results challenge several prevalent beliefs about medicine supply chains in LMICs and lead to hypotheses that require further testing. We also discuss how such testing could improve our understanding of inventory management in private healthcare facilities and aid progress in achieving UHC.

Keywords: Inventory control; Low- and Middle-Income Countries; Urban neighborhoods; Private healthcare; Semi-structured interviews;

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1 Introduction

> *"Nearly 2 billion people have no access to basic medicines, causing a cascade of preventable misery and suffering."*
>
> *—Dr. Margaret Chan, Director-General of WHO, 2007 – 2017*

Universal health coverage (UHC) is an integral part of the United Nations sustainable development goals (UNSDG 2015). Encompassed within UNSDG #3, UHC includes ensuring "...access to safe, effective, quality and affordable essential medicines and vaccines for all" by the year 2030. However, availability of essential medicines remains poor in many low- and middle-income countries (LMICs) (Ewen et al., 2017; WHO, 2013), and this can lead to adverse outcomes for patients as a result of not receiving treatment or treatment being delayed or discontinued (Elbireer et al., 2011; Pasquet et al., 2010; Mendis et al., 2007).

The private sector has an important role to play in achieving UHC in general, and specifically in improving access to essential medicines in LMICs (Frohlicher and Nouwen, 2019; Yadav and Glassman, 2019), where a very significant part of healthcare expenditure is not publicly funded (Chan, 2018). However, studies report that availability targets for essential medicines are far from being met by private healthcare facilities, particularly so in low-income areas. For instance, Cameron et al. (2009) use data from 36 LMICs and calculate that availability of 15 important generic medicines averages only 64.2% for the private sector (38.4% for the public sector).

Other studies too have identified a mismatch between demand and supply in private healthcare facilities (Ewen et al., 2017; WHO, 2013; Ongarora et al., 2019), but there have been few rigorous and data-based analyses of the underlying causes. Causes mentioned in the literature often come with no or limited evidence. Articles that do present data-based analyses usually take a limited perspective – for example, focusing solely on supply issues or skills. This raises the risk of policy-makers jumping to premature conclusions and focusing on ineffectual remedies.

Through an exploratory study, this paper aims to understand the underlying causes of the inventory management problems in these facilities. Having a comprehensive understanding of the underlying causes is key to suggest effective remedies to these problems (Yadav, 2015). Effectively remedying the mismatch between supply and demand in private healthcare facilities is a key development priority. Availability of essential medicines not only impacts "health and well-being", but also progress with other UNSDGs (Tosun and Leininger, 2017). UHC can

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prevent people from being pushed into poverty (UNSDG #1) and social inequality (UNSDG #10). Health and well-being are also closely related to economic growth and ability to work (UNSDG #8).

Our work has two major contributions to the Operations Management literature. First, it helps to advance our understanding of the inventory management practices in private healthcare facilities in resource-limited settings. Since extant literature does not provide an explanation for the observed mismatch between supply and demand, and since findings from the public sector cannot trivially be transferred to the private health facilities (see Section 3), we chose an exploratory research methodology. Based on our analysis, we put forward several propositions that should be tested in future work to further our understanding of inventory management in healthcare facilities in LMICs.

Second, our work probes the validity of mainstream assumptions in retail supply chain management models when applied to resource constrained settings. The rapid penetration of point of sale (POS) data systems and automated inventory management systems has improved inventory replenishment decisions in retail stores in industrialized settings. This has allowed retail inventory managers in the last decade to focus more on 1) store execution (Fisher et al., 2006; Raman et al., 2001), 2) improving replenishment decisions (Cachon and Fisher, 1997; Fisher et al., 2001; Van Donselaar et al., 2010), and 3) understanding supplier-retailer interactions. When applying mainstream retail supply chain management models to health product retailing in resource limited settings, it is not clear which deviations from the accepted modeling constructs are the most significant. We highlight a few such deviations (see Table 4), and believe they would help in selecting the right assumptions (and constraints) in future modelling-based research in this area.

Data for this study was collected through in-depth, on-site interviews with 39 facility managers making inventory decisions in private healthcare facilities located in low-income areas of Nairobi. This sample is also academically interesting for three reasons. First, given the pace of urbanization in LMICs, it represents a context that is becoming increasingly relevant (United Nations, 2018; Management Sciences for Health, 2014). Second, the inventory control decisions taken by the facility managers in our sample are sub-optimal in a peculiar fashion. We observe a low availability of many essential medicines and seemingly excessive quantities of other drugs

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at the same time. Third, as we show, two commonly cited causes of understocking – supply problems and lack of incentives for inventory optimization – are of relatively little importance for our sample (cf. Wagenaar et al., 2014).

Using semi-structured interviews this study performs a deep exploration of inventory management practices at these facilities on the following dimensions: 1) inventory control systems, 2) inventory control skills, 3) time/human resource constraints, 4) budget constraints, 5) motivations for inventory control, and 6) suppliers.

Our results suggest that poor inventory control skills, inadequate systems, limited overage/underage costs, and resource constraints are major causes of the observed mismatch between demand and supply. The last two of these seem to be exacerbated in part by two more fundamental issues: lack of regulation in the prescription drug market, and low barriers to entry. Lack of regulation allows healthcare facilities to substitute freely if they run out of stock of a particular drug, and to prioritize drugs that they have available over those that are most appropriate. Low entry barriers lead to fierce competition, which appears to underlie budget problems and to limit facilities' opportunities to invest in developing skills and systems. Our results also add nuance to several prevalent beliefs about the over- and understocking of essential medicines and corresponding solutions, and have implications for the WHO guidelines on quantifying availability.

In the next sections, we review the related literature (section 3), describe our methodology (section 4) and present our results (section 5). Finally, we discuss our conclusions (section 6) and their implications for policy-makers and others who seek to improve the availability of medicines (section 7).

2 Background

In 2016, we conducted a large descriptive study of private healthcare facilities in Nairobi to investigate medicine pricing (Ongarora et al., 2019), adherence to quality standards (Abuga et al., 2019), and availability of essential medicines. During this study, trained field agents did a physical inventory count at participating facilities to determine whether the items included in the Kenya Essential Medicine List (KEML)[1] were in stock. The field agents also used structured

[1]The KEML is a list of drugs, drawn up by the Kenyan Ministry of Health, that are "considered to be the most efficacious, safe and cost-effective for the relevant conditions" and that are "expected to be routinely available in

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questionnaires[2] to record details of all orders placed during an eight weeks period (October to December 2016) to analyze the ordering process during this period (i.e. the rationale behind ordering decisions and selection of suppliers).

Using data from the physical stock count, we found that, on average, facilities stock only 28% of the items on the KEML[3]. In other words, these private healthcare facilities were understocking essential medicines.

On the other hand, many of these facilities stocked excessive quantities of other drugs. 80% of the items recorded in our baseline stock count were not reordered during the next eight weeks – i.e., facilities held at least eight weeks of stock for 80% of the items. Since lead-times are very short in our study context (see Section 5), and since high stock levels tie up scarce and expensive resources (such as shelf-space and working capital), this is an indicator of overstocking behavior.

The 2016 study was not designed to systematically and holistically assess why facilities have a shortage of many essential drugs and simultaneously stock excessive quantities of non-essential drugs. The next section takes stock of possible explanations for this phenomenon presented in extant literature.

3 Literature review

Grey and academic literature analyzing the over- and understocking of medicines in Nairobi or Kenya is scant. We found two studies that cover private sector facilities as well as public facilities and faith-based health services (FBHS). A study by Health Access International (2018) used data from 82 facilities to show that locally produced products are more affordable and more available in the facilities surveyed. This suggests that cost is a driver of availability, which is understandable from an inventory holding costs perspective. Based on data collected from 108 facilities, a report by the Kenyan Ministry of Medical Services (2009) shows that about two-thirds of public and FBHS facilities lack qualified pharmaceutical personnel, which the authors argue goes hand in hand with stock-outs and inadequate stock-keeping records. Hence qualified personnel appears to be a major driver of availability as well. Three other reports consider only public facilities. Muhia et al. (2017) and Wangu and Osuga (2014) studied the problems

healthcare facilities (at the appropriately designated levels of care)" (Ministry of Health, 2016)

[2]See Appendix C for the full set of questions.

[3]37% of the items stocked by a facility were from the KEML.

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with the availability of medicines in public hospitals. They found the main reasons to include budget shortages, delays due to bureaucracy, distribution problems, lack of skilled personnel, inappropriate selection of medicines, irrational use of medicines, and an inability to accurately estimate requirements. This last issue "led to an overestimation or underestimation of the drugs needed" (Muhia et al., 2017). Budget shortfalls, supplier issues, and lack of skilled personnel were also identified as causes by Muiruri and Mugambi (2017), who studied 94 public healthcare facilities.

To get a more comprehensive view of current knowledge relating to over- and understocking of medicines in LMICs, we also performed a systematic literature review. We searched the PubMed and Web of Science databases for articles written in English within the last 20 years (going back from 1 October 2018), and available in full-text. Search terms (see Appendix A) were chosen so as to find papers relating to 1) medicines, 2) over-/understocking, 3) healthcare facilities, and 4) low-income settings. Two authors independently assessed the abstracts of the resulting 150 articles. An article was included if its aim was to analyze availability of medicines in healthcare facilities in a low- or middle-income setting, and if this analysis was based on data (primary or secondary). We subsequently read the resulting 20 articles to assess 1) the study sample, 2) what the causes of over- and/or understocking (if any) were said to be, and 3) whether data was presented in the paper to support these claims. Table 1 presents our results.

Our review suggests that holistic, data-based analyses of underlying causes are scant, particularly for private facilities. Only 11 papers discuss causes of over- and/or understocking and eight papers present data-based evidence to support this discussion. Only two of the eight papers that analyze the causes – Ngcobo and Kamupira (2017) and Poku et al. (2017) – cover more than one category. None of the papers appears to take a comprehensive view of the causes.

Moreover, none of the papers focus on private facilities. A similar conclusion was also reached by Yadav (2015), who states that "As with the public sector supply chains, there is lack of a systematic diagnostic methodology on why the private sector supply system cannot meet large public health objectives".

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Article	Public/private/NGO	Rural/urban	Causes reported	Data-based?
Bissell et al. (2016)	Public & private	Not reported	None	No
Chalker et al. (2015)	Public	Rural	None	No
Goudge et al. (2009)	Public & private	Rural	None	No
Harding et al. (2014)	Public & NGO	Not reported	Inventory control policies	Yes
Harrison et al. (1998)	Not reported	Rural	None	No
Kibirige et al. (2017a)	Public & private	Not reported	None	No
Kibirige et al. (2017b)	Public & private	Not reported	Skills, budget, supply issues	No
Lauffenburger et al. (2011)	NGO	Rural	Medicine donations	No
Leung et al. (2016)	Public	Rural & urban	Inventory control policies	Yes
Magadzire et al. (2014)	Public	Rural & urban	Supply issues	Yes
Magadzire et al. (2017)	Public	Urban	Supply issues	Yes
Musinguzi et al. (2015)	Public & private	Not reported	None	No
Ngcobo and Kamupira (2017)	Public	Not reported	Skills, time, budget, supply issues, storage space	Yes
Nilseng et al. (2014)	Public	Rural	Supply issues	Yes
Penfold et al. (2013)	Public	Rural	Supply issues	Yes
Poku et al. (2017)	Public & NGO	Urban	Supply issues, budget	Yes
Poyer et al. (2015)	Public & private	Rural & urban	None	No
Robertson et al. (2009)	Public & private & NGO	Rural & urban	None	No
Wiernikowski et al. (2014)	Public & private	Not reported	Supply issues	No
Wirtz et al. (2016)	Public & private		None	No

Table 1: Causes of over- and understocking mentioned in the literature.

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The main supplier-related causes of understocking reported include supplier stock-outs, communication problems with suppliers, and insufficient resources to ensure reliable and frequent distribution. Inadequate inventory management policies (or no policy at all) and non-adherence to such policies are also seen as other reasons for why stock-outs occur. Harding et al. (2014) show that they occur less in facilities with a reorder point, and Leung et al. (2016) show, in a simulation study, that they occured more frequently when there are inventory policies that are based on false assumptions regarding lead-times and seasonality of demand. Other causes of stock-outs include limited fridge capacity (for vaccines) and lack of pharmacy assistants (Ngcobo and Kamupira, 2017). Among the papers we considered, Lauffenburger et al. (2011) is the only one reporting on overstocking – the authors mention medicine donations as the main reason for overstocking.

This paper endeavors to fill the gap in our understanding of inventory management in private healthcare facilities in LMICs. Through a comprehensive qualitative analysis of possible reasons for the observed mismatch between demand and supply in these facilities, we test several widely held beliefs about medicine supply chains in LMICs, and either debunk or add nuance to them. We also proposes several hypotheses that could help spur much-needed future work on private healthcare facilities in LMICs.

4 Sample Selection and Interview Methodology

In the previous section, we show that extant literature is silent on why private healthcare facilities in Nairobi have a shortage of many essential durgs, while simultaneously stocking excessive quantities of non-essential drugs. The primary objective of this study is to fill this gap in our understanding by asking, *"Why is there a mismatch between demand and supply?"*. We note that this is a "why" type question, as opposed to a "what" type (Yin, 2003). In other words, we do not use data to test an existing theory or a pre-formulated hypothesis claiming a causal relationship. Our research is exploratory, and the exact nature of the theory that would shape up from this study is not known beforehand. When asking such exploratory question, taking the context into consideration is crucial, and qualitative studies are seen as most appropriate (Baxter et al., 2008).

In order to investigate the observed mismatch between demand and supply, we went back to

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the same facilities that participated in the 2016 study. The population from which these facilities were drawn consisted of 381 private healthcare facilities in Nairobi County that were affiliated to PharmAccess Foundation Kenya. Pharmaccess Foundation Kenya is an NGO providing healthcare financing, digital health and healthcare supply solutions to make quality healthcare accessible and affordable to Kenyans. Ninety-one facilities met the inclusion criteria of having a valid license and selling drugs to patients at the bottom of the wealth pyramid. Facilities in low-income areas in Kibera were excluded because in that locality there are several NGOs providing free healthcare services, which may impact competition and the way those private facilities operate. Forty-eight facilities were subsequently selected for inclusion, considering 1) accessibility, 2) distribution of facilities across parts of the county, and 3) representation of different types of facilities. This selection was done by three co-authors with expert knowledge of the local market. Forty-five of the facilities consented to participate in the 2016 phase of the study.

To recruit facilities for the 2018 study, we approached the 45 facility managers in November 2018 and told them about the purpose, benefits, and risks of the study. Facility managers agreeing to participate were asked to provide written consent to their data being used for research purposes, but they were allowed to withdraw from the study at any time. No financial incentives were offered to the facilities to participate. There were also no negative consequences to facilities that chose to not participate. However, during the recruitment phase facilities were promised a post-analysis town hall meeting where results from our studies would be shared.

Though our results have a reasonable degree of external validity for the population of urban private health facilities in low-income areas in LMICs, we stress that the primary intent of qualitiative research "is not to generalize to a population, but to develop an in-depth exploration of a central phenomenon" (p. 206, Creswell, 2009) (we revisit this in Section 7). The aim is to learn from "information rich" interviewees who can best help understanding the observed phenomenon – over- and understocking in private healthcare facilities – and to achieve "information saturation". This is defined as the point when "no new information or themes are observed" with each subsequent interview (Guest et al., 2006). Our 2018 sample forms a unique set of such "information rich" interviewees. Guidelines suggest that semi-structured interviews require a sample between 25 and 30 interviewees to achieve information saturation (Symon and

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Cassell, 2012; Lewis, 2015). Following these guidelines, we believed at the beginning of the 2018 study that our sample was sufficiently large, despite the withdrawal of few facilities. This belief was confirmed during the actual field work, when interviewers reported information saturation approximately after the first 20 interviews.

The 2016 sample consists of 18 clinics, seven hospitals, two health centers, four medical centers, two nursing and maternity homes, and twelve pharmacies. We refer the reader to (Ongarora et al., 2019) for details on respondents and additional facility characteristics. Of the 45 facilities that took part in the 2016 phase, 39 did so again in the 2018 phase: one hospital and three pharmacies declined to participate, and two clinics had by then closed shop. Our 2018 sample therefore consists of 16 clinics, six hospitals, two health centers, four medical centers, two nursing and maternity homes, and nine pharmacies. Interviewees had on average 11.3 years of work experience in the pharmaceutical sector. Among them were nine clinical officers, three doctors, 11 nurses, nine pharmaceutical technologists, and six pharmacists (see Appendix B for the pharmaceutical terminology).

To put together a questionnaire for this second survey, we followed a commonly accepted methodology, formalized by Kallio et al. (2016). In Appendix D we present a detailed review of how our process follows the steps laid out by Kallio et al. (2016).

We chose a semi-structured interview format for our study because they are considered to be "flexible, accessible, intelligible and capable of disclosing important and often hidden facets of human and organizational behavior" (Qu and Dumay, 2011). Most importantly, semi-structured interviews enable interviewees to provide responses in their own terms and in the way that they think and use language. This often makes them the most effective and convenient means of gathering information for an exploratory descriptive study (Kvale and Brinkmann, 2009).

Based on causes suggested in the literature, we formulated a preliminary semi-structured interview guide. It was specifically designed to assess each facility's 1) inventory control systems, 2) inventory control skills, 3) time/human resource constraints, 4) budget constraints, 5) motivations for inventory control, and 6) relationship/issues with suppliers. Several open questions were included to capture potential causes not covered by these six categories. The interview guide was reviewed by each author and two other academics with advanced knowledge on the topic. We then proceeded to pilot test the guide during 3 facility visits in August 2018 and

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finalized it afterwards (e.g., we aligned the jargon with that of the interviewees). We present the final version of the interview guide in Appendix E.

4.1 Interview Analysis

In the spirit of semi-structured interviews, the questionnaire was designed to prompt the interviewees to give as full an answer as possible to questions, which might lead the discussion to other tangentially related issues. Interviewers also spent significant amount of time at each facility taking pictures and noting down observations about the facilities. In most cases, the facilities were run by just one or two people – thus, inventory management is just one aspect among many things they need to look after to keep the facility running. Several times during the interviews, the interviewees excused themselves to attend to clients.

Each interview included at least two interviewers: The third author was present in every interview to ensure consistency and was accompanied by a Nairobi local. On average, the interviews took 83 minutes. Interviews were taped and later transcribed by an independent agency.

The transcripts were analyzed using NVivo 12 text analysis software. Before we began the text analysis, we read through all the transcripts end-to-end, to familiarize ourselves with the full content of our interviews. In the first stage of the text analysis, parts of the interview that provided direct answers to the pre-planned questions (see Appendix E) were identified and tagged to the appropriate question.

For instance, question 14 in the interview questionnaire asks interviewees, "Why is it important for you to have drugs on stock?". In our interviews, this question is meant to explore the interviewee's motivations and incentives for stock-keeping and inventory management. Since our interviews were semi-structured, the conversation could touch upon motivations and incentives several times throughout the interview. All parts of the interview in which the interviewee discussed motivations and incentives to stock drugs were labeled "Motivations".

In the next stage of the analysis, we collated all the parts of the transcripts associated with each label and read through them again. We then assigned further lower-level tags (sub-labels) to the collated text. For instance, several possible scenarios were identified as answers to the question, "What happens when the drug a client needs is out of stock?". After going through all

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the answers (across all the transcripts) to that question, we created a list of mutually exclusive and cumulatively exhaustive list of sub-labels associated with each scenario. Whenever an interviewee from a particular facility mentioned a certain scenario, the corresponding sub-label was attached to that part of the transcript.

Also, when a discussion topic emerged that was not covered in the questionnaire, a new label was created for it, and all the relevant parts were tagged under that topic. For instance, it can be seen in Appendix E that the questionnaire included no mention of "regulatory issues" as a topic. However, several interviewees brought up issues related to government regulations when discussing their inventory management practices.

Finally, to perform an objective and quantitative analysis, we formulated several variables, which took a binary value of 0 or 1 based on each facility's interview. One example is, "At least one of the facility's suppliers currently offers a line of credit". The definitions of these variables and their objectivity was discussed by the co-authors and iteratively refined until consensus was reached. All the statistics reported from the interviews are based on such objective variables (see Table 3 for a list of variables).

Upon completion of our analyses, we invited all the participants to a workshop at which we presented and discussed the results (see Section 4.2). Nineteen participants attended. The workshop allowed us to "member-check" the results and hypotheses presented in the following sections.

Ethical approval for the study was granted by University of Nairobi-Kenyatta National Hospital Ethics and Research Committee (UoN-KNH ERC) under approval number P549/07/2016.

4.2 Insights Dissemination Workshop

In August 2019, after we had completed the bulk of the analysis for this paper, we returned to Nairobi and held a workshop for our interviewees. In this workshop, we shared insights from our analysis with healthcare facility managers, and corroborated our thought process in this paper. The facility managers also participated in a root cause analysis using the "5 times *why*" approach (Ohno, 1988). We will refer to insights from this workshop in our discussion (Section 6) and conclusions (Section 7)

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5 Results

This section presents our findings on causes of over- and understocking related to 1) inventory control systems, 2) inventory control skills, 3) time/human resource constraints, 4) budget constraints, 5) motivations for inventory control, and 6) suppliers.

Inventory control systems. Mismatches between demand and supply (which lead to over- and understocking) can be controlled by adopting an adequate *inventory control system*, which we define as the tools, policies, and processes used to make ordering decisions. Inventory control can be enhanced by keeping records of stock purchased, received and sold, and using them for decision-making. This holds irrespective of whether the system is manual or electronic.

Our data reveals that the facilities have given little consideration to inventory control systems, which may partly explain the over- and understocking. Interestingly, while 95% of the facilities (37 out of 39) are familiar with the concept of a "reorder point", hardly any have formulated an explicit policy stating exactly what the reorder points should be. We find similar patterns for order quantities and review periods. Instead, decisions are largely based on intuition and experience.

***Interviewer**: Are you doing any calculation or just your experience?*

***Respondent**: It is through experience. No calculations.*

***Interviewer**: Why did you decide to purchase 50 and not 40?*

***Respondent**: It is not scientific. It's not.*

***Interviewer:** How then do you determine how much to buy?*

***Respondent**: I don't have that. I'm not able to tell how much to buy. I just buy.*

Some facility managers make a physical check of inventory levels at the point they decide to place an order, and decide to buy new supplies of medicines that they deem to be most urgently

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needed. Others make a list of medicines for which they judge the current inventory levels to be too low, and this list is then periodically transformed into an order. Exactly when the order is placed is often an opportunistic decision, depending on time and available budget and the urgency of re-supply (e.g., because a patient is waiting for a drug). The quantity to be ordered is sometimes determined at the last moment.

Facilities also make little use of historical data. About 77% of the facilities (30 out of 39) reported keeping sales records and 79% (31 out of 39) reported keeping historical purchase data. Even though 31% of the facilities (12 out of 39) had an IT system for inventory management, only 10% of them (4 out of 39) reported using data for decision-making. Facilities that do so followed a hybrid approach. They use historical data to extract demand rates or to obtain a recommended order quantity and combine this with their personal judgement when making ordering decisions.

***Interviewer**: Now if your intuition tells you buy 20 but the computer tells you "buy 10", will you buy 20 or will you trust the computer?*

***Respondent**: Some of the things in the computer I can go with them some of them I will not go with them.*

Use of manual or IT tools for inventory control is also scarce in our sample. Forty-four percent of the facilities (17 out of 39) had no tools to keep track of their inventory whatsoever – neither manual nor IT based. Twenty-six percent (10 out of 39) had a bin-card system in place. However, based on the subjective judgement and interview-notes of the interviewers[4], nine out of the 10 facilities with a bin-card system (90%) are not using it properly or consistently.

***Interviewer**: Do you think your bin card system is being used well?*

***Respondent**: For now I cannot say they are well in place*

***Interviewer**: What do you think are the main problems with the bin card system?*

***Respondent**: I think they are not up to date*

[4]The interviewers discussed their observations after each interview and formed a joint judgement.

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The same applies to IT tools used for inventory control. Many were not fit for purpose, underutilized, or not consistently used. According to the interviewers' subjective judgement and their interview-notes, in only 42% (5 out of 12) of the facilities the IT system could be considered "adequate" for the purpose and used it consistently. Twenty-five percent of the facilities (three out of 12) had an IT system that was not fit for the purpose of inventory management in the context of that facility. Another 25% (3 out of 12) facilities had an IT system but did not use it consistently; and 8% (one out of 12) had a system but did not use it at all. These observations raise questions about the cost-effectiveness of IT solutions for inventory control in the kinds of facilities represented by our sample. We return to this discussion in section 7.

Inventory control skills. We define inventory control skills as the capability to set adequate system parameters based on data or intuition, and to execute and adhere to policies and processes. Setting adequate system parameters is a well-studied aspect of inventory management. The following fundamental factors typically need to be considered when determining reorder points: 1) lead-time, 2) rate and variability of consumption, 3) holding costs, and 4) stock-out costs. Factors to be considered when determining order quantities include: 1) holding costs, 2) the rate of consumption, and 3) ordering costs, both fixed and variable (cf. Axsäter, 2015).

When asked to enumerate the factors that are considered when determining reorder points and order quantities, 97% of the facilities (38 out of 39) mention *rate of consumption*, but only 13% mention *lead-time*. Holding and shortage costs were also rarely mentioned explicitly. Most facilities order "when they are almost out of stock".

***Interviewer**: Okay then, how do you decide this is the moment to buy?*

***Respondent**: When we don't have. When we are finishing. When we are about to finish*

***Interviewer**: (...) how do you decide now it's time to order?*

***Respondent**: (...) when I see the ones that I have are almost getting finished*

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Though this decision rule is loosely specified, choosing to order when stock levels are low appears to be rational. This is because lead-times are small and stock-out costs tend to be low; we discuss this in more detail later. Hence, though these factors were rarely mentioned explicitly, they may have been implicitly incorporated.

On the other hand, inventory management theory stresses that the stock-out rate is *a choice* one can make by setting the reorder point. However, this link between reorder points and stock-outs appears to be poorly understood and often not even acknowledged by our interviewees. They primarily mention external factors as causes: "high consumption by patients" in 69% of cases and "supplier shortage" in 20% of cases. Acknowledging that one "ordered too late", which is a different way of saying that one's reorder point was too low, occurred in only 3% of cases. This may exacerbate understocking.

We also asked interviewees why specific order quantities were chosen. While 46% of the facilities (18 out of 39) are familiar with the concept of "order quantity", they were unable to offer a satisfactory rationale for the specific order quantities chosen.

When discussing order quantities, interviewees did not mention ordering costs; this is understandable since all of them indicated that they do not pay any delivery fees. However, it is surprising that holding costs are rarely accounted for. Only 14% mention that budget, for which holding costs could be seen as an indicator, affects quantity decisions. Instead, 64% report that order quantities are based on consumption rates – more demand means larger order quantities – which can be far from optimal. Facilities also made too much use of stock investment heuristics. This general failure to consider holding costs may at least partly explain the overstocking phenomenon.

Our data set does not allow us to assess the second aspect of inventory control skills, the capability to devise and adhere to policies and processes, since none of the facilities uses explicit inventory control policies.

Time/human resource constraints. Time and manpower constraints could also lead to mismanagement of inventory and hence over- or understocking. As a facility increases in size, it often becomes infeasible for a single person to manage. Of the facilities in our sample, 61.5% (24 out of 39) report that having more time would improve their inventory management.

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***Interviewer**: So how would time make records available?*

***Respondent**: In most cases you forgot to put it down. If you have enough time you can go through the shelves, and you realize you forgot to list this down.*

***Interviewer**: Why would it improve if you had more time?*

***Respondent**: Because I could check* [SKUs] *one by one. Now I check few (...)*

***Interviewer**: Do you have the feeling that, if you had more time for stock management, it would go better?*

***Respondent**: In terms of checking my stock-outs, in terms of checking my expiring drugs, checking my overstocking, yeah, of course.*

We believe that some of the facilities would benefit from hiring staff to take over mundane tasks, as this would free up the manager's time to deal with other issues. However, they report that it is hard to find trustworthy personnel. Pilferage on the premises had been experienced by 69.2% of facilities (27 out of 39), and in most cases this was by a staff member.

***Interviewer**: Finding people that you can trust for your business is an issue?*

***Respondent**: Very challenging. In fact I would almost say "No, you cannot leave someone to stand-in for your business as you would yourself"*

***Respondent**: Yeah, some staff take. They swallow. They don't record. Or somebody can sell, does not record, so they pocket money. So at the end of the day you get losses.*

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Budget constraints. Cash-flow problems appear to be one of the biggest constraints for facilities. Budget constraints can evidently lead to stock-outs and sub-optimal order quantities. All of the facilities (39 out of 39) consider budget to be a constraint on inventory management.

***Interviewer**: OK, then how do you determine the quantities to order at the beginning of the month?*

***Respondent**: Actually I call, placing the order. And then I keep asking them "What's my total?" "What's my total?"(...). Then, when I get close to the amount that I want to spend, then I start cutting down on the quantities.*

***Interviewer**: ... why did they go out of stock briefly?*

***Respondent**: There was a challenge there was due to financial constraints*

***Interviewer**: Financial constraints?*

Respondent: [When there is a] *failure to pay promptly for a particular supplier they discontinue supplying you. Then you try to reduce your stocks and order levels.*

We did not explicitly ask interviewees to state the reasons for their financial constraints. However, during the course of a few interviews, delayed payments by the National Hospital Insurance Fund (NHIF) came out as one reason.

***Respondent**: This also can be due to the fact that NHIF usually pays in a period of one to three months, the capitation payment. ... So you see* [there is a] *deficit of funds because of the long time NHIF takes to pay the money.*

The reported level of cash-flow constraints is surprising considering that 69% of facilities (27 out of 39) are offered a line of credit by at least one of their suppliers. Not all interviewees had a positive view of such credit options. Interviewees said that they did not like to be "indebted" to their suppliers. This suggests that facility managers lack the necessary understanding of

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how risk-sharing contracts can benefit them. On the other hand, a few interviewees expressed concerns that having a line of credit leads them to purchase more than is necessary, because the transfer of money for goods purchased is not immediately salient. This suggests that, similar to inventory decisions, financial decisions are largely driven by "intuition" and gut feelings.

***Interviewer**: Do you think credit is a good thing?*

***Respondent**: Yes and no. Yes because it gives you leeway to sell and to pay, no because you are on credit. You are on debt.*

***Interviewer**: Do you think credit is a good thing?*

***Respondent**: The advantage is it can help you when you don't have cash (...) and the disadvantage is (...) you can order more and find difficulty in paying.*

Motivations for inventory control. Having facility managers with the "right" motivations is an important precondition for adequate performance. Two motivations that are important in terms of avoiding over- and understocking are: minimizing costs (i.e., holding costs, ordering costs, and stock-out costs) and providing the best possible service to patients. We assessed the interviewees' motivations by asking them to describe why they consider it important to have drugs in stock, to describe their assortment selection criteria, and what their considerations are when determining which drugs to keep in stock and what quantities to order.

The findings from the 2016 phase of the study indicate that facilities assign on average 65% weight to demand and 28% to whether the item is on the KEML list when determining whether or not to stock a drug. They thus have a particular incentive to stock items for which there is high demand, but also to stock medicines that are essential for their clients.

***Interviewer**: Why is it important for you to have drugs on your shelves? To have drugs in stock?*

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Respondent: [chuckles] *It is the major reason why I am here. Like I am open so that I can attend to the patient (...) when they need medicine. So that's why I am here. So they have to get help from here.*

Interviewer: *Why is it important for you to have drugs on your shelves? To have drugs in stock?*

Respondent: *Because we are in business, we are not selling shelves here.*

As stressed by *newsvendor theory* (Axsäter, 2015), incentives to avoid understocking are stronger when the shortage costs – the opportunity costs of not being able to meet a demand from stock – are high. Interestingly, 74% of the facilities (29 out of 39) indicate that they can backlog demand, suggesting that shortage costs are rather low. If they run out of a certain medicine, they can order it for the client and ask him/her to pick it up at a later time. A few facilities report that they request partial payment to ensure pickup.

Interviewer: *If you are to make an order to your main supplier, it will take time. Will the patient wait?*

Respondent: *Yes, some people even give a down payment to ensure that.*

Seventy-two percent of the facilities (28 out of 39) report that they offer the client a substitute drug when they experience a stock-out. Fifty-six percent of the facilities (22 out of 39) report that they turn to neighboring facilities to "borrow" the necessary drugs.

Interviewer: *(...) you go looking in your shelves and you see this is not available. What would be the first thing that you will check?*

Respondent: *An alternative.*

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***Respondent**: I would borrow from the clinic ahead or some friends across the road.*

***Interviewer**: Borrow means: they give and you return the item or you pay for it?*

***Respondent**: No, usually we borrow from each other. You make an order and you return the item.*

As a last resort, facilities can also give out a prescription, which the clients can use to purchase the medicine at a different facility. However, this would mean a loss of business for the facility giving out the prescription. Forty-six percent of the facilities (18 out of 39) report that they would recommend a specific trusted pharmacy to the client when giving out a prescription.

***Respondent**: (...) people should have this trust in us, "if I go there I will get these services, I will get the drugs I need"*

The fact that delivery lead-times are short (see below) suggests that stock-outs have relatively few health implications for patients in terms of their treatment being either delayed or discontinued. Hence, though facilities appear to have a strong drive to minimize costs and maximize service to clients, there is not a strong incentive to avoid stock-outs.

We now turn our attention to assessing motivations to minimize inventory *holding costs*. We provide a rough estimate of these costs based on the value (procurement costs) of the inventories recorded during our 2016 baseline assessment. As we did not collect procurement costs directly, we estimate them by applying a 33% sales margin to the retail prices (the average margin on products *procured* during the eight-week study period). The majority of the facilities (35 out of 45) could not provide any detailed information on retail prices, which were therefore excluded from this analysis. For the remaining 10 facilities, the average inventory value is 229,454 Kenyan Shillings (KShs), which is about 2,200 US dollars. Applying the common rule of thumb that annual holding costs are 25% of the average inventory value (Azzi et al., 2014), we estimate that average annual holding costs equal 57,363 KShs (555 dollars), which is a considerable sum in the context of our study. It is therefore surprising that substantial overstocking occurs, despite the observation that facilities have a strong drive to minimize costs.

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Our semi-structured interviews also shed light on a worrying practice we call *demand shaping*. Facilities that engage in this practice prescribe and push drugs that are available in stock rather than those that are most appropriate for the clients' needs. This lack of regulatory oversight incentivizes facilities to stock and prescribe broad-spectrum and unnecessarily strong drugs for a wide range of patient complaints. Such practices could potentially lead to an increase in anti-microbial resistance.

***Interviewer**: So what do you do if you notice that a certain drug is about to expire in the next three to four months?*

***Respondent**: ...we inform the clinician, "this drug is almost expiring so kindly try and see how I can move." When they get a case, they prioritize that drug.*

Suppliers. Stock-outs can evidently be affected by access to suppliers, availability of medicines at the suppliers, and lead-times. Almost all interviewees in the 2018 phase of the study (97% of the facilities: 38 out of 39) indicated that they had encountered stock-outs at their main suppliers. Furthermore, for 20% of the stock-outs that occurred during the 2016 phase, respondents mentioned "supplier shortage" as a reason.

***Interviewer**: (...) so do you sometimes experience that you call the supplier and they say I don't have this drug, they are out of stock. Does that happen?*

***Respondent**: Yes*

***Interviewer**: Is it frequent?*

***Respondent**: No*

However, supplier-related causes are only a minor issue for several reasons. First, the number of suppliers serving Nairobi is high. In both data-gathering phases, we asked facilities to list their suppliers, which yielded 112 unique names. Seventy-seven percent of the facilities (30 out of 39) have at least three suppliers from whom they regularly source medicines, and 90% of the

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facilities (35 out of 39) report reaching out to other suppliers when their main supplier runs out of stock for a certain drug.

Second, the vast majority of orders (91%) are delivered the same day. Delivery time is so short for most facilities that only 13% of them (5 out of 39) factor delivery lead-time into their ordering decisions and only 23% of the facilities (9 out of 39) consider maintaining a buffer stock to hedge against unexpected delivery delays. Third, in 84% of cases, facilities received all the medicines they ordered.

Suppliers can also cause overstocking by incentivizing facilities to purchase large quantities through promotions and offers. Almost all the facilities report that they are approached by their suppliers with such offers. However, 79.5% of the facilities (31 out of 39) report that they do not budge from their order quantity decisions on the basis of such offers.

***Interviewer**: Do you often fall for these offers?*

***Respondent**: It depends. (...) If we need it, yeah, we get it.*

However, some respondents may be tempted by these offers.

***Interviewer**:* [The drug] *Penamox. Why do you have so much?*

***Respondent**: You know at times you get a good offer(...) you buy in bulk only to realize that it doesn't move as fast.*

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Summary Table 2 summarizes the key insights generated from our interviews for the six categories of causes we considered. Table 3 contains a list of questions with binary answers used to present the statistics in Section 5.

Inventory control systems
Poor use of historical data
Ordering decisions based on intuition and experience
Improper and inconsistent use of inventory control tools
Inventory control skills
Poor understanding of the link between reorder points and stock-outs
Limited rationale for order quantity decisions
Holding costs largely ignored
Time/human resource constraints
Facility managers overwhelmed by amount of work
Hard to find trustworthy personnel
Budget constraints
Ordering decisions heavily constrained by budget
Line of credit offered by most suppliers
Dislike of using suppliers' credit facilities
Motivations for inventory control
Facilities clearly profit-driven
Financial incentives to stock high-demand products
Limited under-stocking costs due to backlogging and informal inventory pooling networks
Incentives to avoid over-stocking potentially limited due to demand shaping
Relationship/issues with suppliers
Short lead-times for supply
Large number of suppliers
Most facilities have at least three different suppliers
Stock-outs at main suppliers do occur
Most facilities reach out to other suppliers when this happens

Table 2: Key insights from our interviews, by category.

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Question	+ve responses
Does the facility have at least three suppliers?	77% (30 out of 39)
Does the facility reach out to other suppliers when their main supplier runs out of certain medication?	90% (35 out of 39)
Does the facility have at least one supplier that offers it a line of credit?	69% (27 out of 39)
Does the facility consider the suppliers usual lead-time when choosing to do business with them?	13% (5 out of 39)
Does the facility only accept supplier offers after careful consideration?	79% (31 out of 35)
Is the facility manager familiar with the concept of reorder point?	95% (37 out of 39)
Is the faculty manager familiar with the concept of reorder quantity?	46% (18 out of 39)
Does the facility manager realize that seasonality impacts inventory decisions?	100% (39 out of 39)
Does the facility manager mention that ordering decisions take into account the *buffer stock levels* ?	23% (9 out of 39)
Does the facility manager mention that ordering decisions take into account the *rate of consumption* ?	97% (38 out of 39)
Does the facility manager mention that ordering decisions take into account the *delivery lead-time* ?	13% (5 out of 39)
Does the facility manager use data to make inventory decisions?	10% (4 out of 39)
Does the facility report tracking historical order data (invoices, for instance)?	79% (31 out of 39)
Does the facility report tracking historical demand data (sales, for instance)?	77% (30 out of 39)

Table 3: Overview of responses

6 Discussion

Streamlining medicine supply chains in LMICs is a necessary condition for achieving universal health coverage, as targeted in the United Nations Sustainable Development Goals 2030. Very little healthcare provision in these countries is publicly funded; most patients pay for medical expenses out of their own pocket (Chan, 2018). Thus, the private sector has an important role to play in achieving these goals.

Our 2016 field study revealed that private healthcare facilities in Nairobi's low-income areas stock seemingly excessive quantities of many drugs. At the same time, they have no stock of many essential medicines. Since these facilities are privately owned, they are clearly profit-driven and have complete control over their inventory management choices. Thus, such a pronounced mismatch between demand and supply would suggest that deeper factors are driving

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this behavior.

In the semi-structured interviews conducted in 2018, we investigated this over- and understocking of medicines and assessed each facility's 1) inventory control systems, 2) inventory control skills, 3) time/human resource constraints, 4) budget constraints, 5) motivations for inventory control, and 6) relationship/issues with suppliers (see Section 5). In this section we will discuss the causes related to each of these categories, elaborate on the relationships between them, and interpret them in relation to our context. We put forward several propositions that can guide future research into this issue.

As we show in Section 5, supply problems appear to play a negligible role in causing over- or understocking of medicines. Even when facilities occasionally encounter stock-outs at their primary supplier, they have access to an abundance of alternative suppliers. Given that availability at the supplier-level is high in many urban contexts in LMICs (Yadav and Glassman, 2019), we hypothesize that:

> **Proposition 1.** *Low availability of medicines in private healthcare facilities located in urban neighborhoods in LMICs is primarily driven by facility-level issues rather than by unavailability at the supplier-level.*

Our analysis shows that these facility-level issues are weak systems, lack of skills, budget constraints and, to some extent, time/HR issues. Section 5 highlights how facilities do not take advantage of historical orders and sales data to make inventory management decisions but rely too much on their intuition and experience. Facility managers also report inconsistent and ineffective use of inventory control tools. Moreover, many facility managers do not have sufficient understanding of the link between reorder points, order quantities and stock-outs. We also find that almost all facilities face budget constraints when implementing their inventory management strategy. These findings coincide with the subjective judgement of the interviewers, who sat together after each interview to assess the scale of the problem faced by that facility in each of the six categories shown in Fig. 1.

Our analysis also finds that these facility-level issues are not due to a lack of motivation to ensure inventory control but may be driven by insufficient incentive to optimize those control parameters. That is, we find that facilities are motivated to maximize both profits and patient

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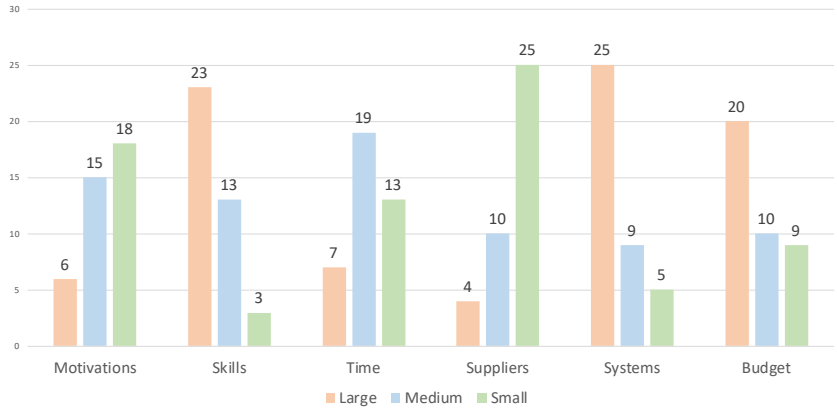


Figure 1: Interviewers' subjective judgement of the scale of the problem. Note that motivations and suppliers are usually perceived as minor problems, time as a moderate problem, and skills, systems, and budget as major problems.

satisfaction (and therefore aim to stock items in high demand; cf. Section 5), but they do not have a strong incentive to maintain high service levels. Facilities can easily "borrow' inventory from a neighbor or backlog their demand. This is facilitated by the very short lead-times and the large number of suppliers in the market, which drastically reduces the facilities' underage costs and the impact of stock-outs on their clients. In the 2016 study, when asked to score the importance of attributes by which patients purchase drugs, facility managers assigned only 14% of the total weight to availability (as against 34% to price, 24% to quality, and 16% to efficacy). This leads us to hypothesize that:

> **Proposition 2.** *Private healthcare facilities in LMICs choose not to stock many essential medicines because they perceive the underage costs for those medicines to be so low that there is no financial rationale for stocking them.*

On the other hand, facilities report substituting prescribed drugs with what they have in-stock. This goes against regulations, but there is little or no enforcement. A consequence is that overage costs are limited, so overstocking is not considered a problem.

> **Proposition 3.** *Holding too much stock of a certain medicine makes it more likely that the medicine will be prescribed to patients, even though there may be better alternatives.*

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Fig. 2 summarizes the six proximate causes, which we classify into three broad classes: 1) *structural causes* – external issues that affect the incentives to avoid over- and understocking, 2) *managerial causes* – internal issues relating to the way inventories are being managed, and 3) *resource limitations* (human and monetary).

It is important to note that these causes are not independent of each other. We hypothesize that there are strong linkages between them and deeper factors that cause them to spread.[5] We next explore these possible linkages (see the dashed arrows between causes in Fig. 2), acknowledging that there is a need for further research to substantiate them.

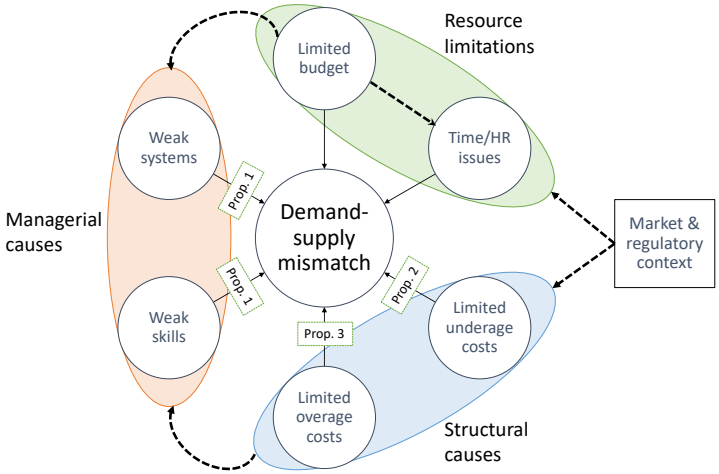


Figure 2: Causes of demand-supply mismatches in private healthcare facilities in Nairobi. Solid arrows indicate a direct link between a cause and demand-supply mismatch. Dashed arrows indicate our hypothesized, exacerbating linkages between causes.

First, budget constraints not only directly influence stock-outs but also aggravate other problems, potentially leading to a downward spiral in inventory management performance. During a feedback and dissemination workshop for interviewees, conducted in 2019, we let participants perform a root cause analysis using the "5 times *why*" approach (Ohno, 1988). Budget came up as an issue that influenced 1) skills (no budget for inventory management training), 2) systems (no budget for professionalizing inventory control systems), and 3) time/HR (no budget to hire extra personnel, and low budget for salaries leads to high staff turnover).

Second, we hypothesize that structural issues also influence managerial factors. Zomerdijk and De Vries (2003) note that, in many cases, poor inventory management performance triggers

[5]Though our interviews did not capture these factors systematically, they surfaced regularly. We were able to piece together a framework on the basis of our general understanding of the market structure and regulatory landscape, insights obtained during the interviews, and a dissemination workshop for interviewees held in 2019.

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a "diagnosis" of the situation. However, due to the limited overage and underage costs in our context, that did not seem to happen in many facilities. This keeps the facilities from investing in improving their skills or implementing efficient inventory control systems.

Finally, these proximate causes seem to be driven by deeper issues that are linked to the market and regulatory context. In Nairobi, the client population is poor and the density of facilities is high[6]: 1.24 per square kilometer (Baker et al., 2014). This causes catchment areas and sales margins to be small, demand to be low, and demand variance to be high. Low demand limits their budgets and opportunities to invest and professionalize, exacerbating the managerial issues.[7] Furthermore, market conditions and regulation also influence the structural causes. As we note in Propositions 2 and 3, market conditions and lack of proper enforcement allows demand shaping and informal inventory-pooling networks to develop, softening both the economic impact of the demand-supply mismatch and its impact on patients.

We note that our findings are cannot be trivially generalized from findings on the public sector healthcare facilities in LMICs. For instance, "diffuse accountability" and "lack of incentives for supply chain staff" was identified as a root cause for stock-outs in public sector healthcare facilities. Whereas, our interview results suggest that motivations and incentives are not a concern for most of the facilities in our sample (Yadav, 2015). Similarly "long resupply intervals" was identified as another root cause for mismatch between supply and demand in public healthcare facilities. However, our interviews reveal that the delivery lead time for most facilities in our sample is less than one day – in fact, it is often in the order of hours.

Our analysis also identifies several differences between mainstream retail inventory management assumptions (typically based on industrialized settings) and the realities of health product retailing in resource limited settings. In Table. 4 we highlight these differences. We hope that the propositions laid out in this paper, together with the differences in Table. 4, would help in selecting the right assumptions (and constraints) in future modelling-based research in this area.

Even though the primary purpose of our study is exploration, we believe our insights have a reasonable degree of external validity. First, while trying to achieve information saturation,

[6]During our interviews, many facilities (51%; 20 out of 39) acknowledged that they face high levels of competition.

[7]During our feedback workshop, 60% of respondents (9 out of 15) agreed with the statement, "There are too many facilities that sell medicines in my area. If there were less competition, I would be able to further professionalize my business."

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we also endeavored to select a sample that is representative of the population of urban private healthcare facilities in Nairobi. This selection was done by three co-authors with expert knowledge of the local market. Second, our study setting is similar other urban settings in LMICs, i.e. markets with poor clients where there is a high density of severely resource-constrained facilities and suppliers, and also facility managers who have not been well-trained in inventory management(Yadav and Glassman, 2019). In the following section we discuss the implications of these conclusions and potential avenues for future research.

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Area	Mainstream retail inventory management assumptions	Health product retailing in resource limited settings
Observing stock-outs	• Shelf-level stock-outs and store-level stock-outs are often very different. • Store-level stock-outs are identified through inventory control systems. • Shelf-level stock-outs are imputed from POS data.	• Facilities are small and have all inventory directly behind the counter. • The owner/manager typically carries out all transactions. • In most cases, stock-outs are directly observed by the store manager.
Nature of retail competition	• Intense competition, extensive price comparison and substantial stock-out costs (cf. Anderson et al., 2006)	• Facility managers "borrow" stock from their competitors in case of a stock-out. This suggests that retailers are not necessarily competing on availability. Quality/efficacy and price appear to be the major drivers of competition amongst facilities.
Credit issues	• Vendor managed inventory (Çetinkaya and Lee, 2000; Yao et al., 2007) and consignment stock (Valentini and Zavanella, 2003) have led to retail ordering decisions being less constrained by working capital	• Working capital constraints may be the main factor impacting ordering decision. • Credit constraints also impact the number of suppliers in each category.
Inventory control, forecasting skills	• Automated store ordering systems have reduced the need for these skills at each store. Current focus is on understanding when store managers should make manual over-rides to automated decisions (cf. Kesavan and Kushwaha, 2020)	• Ordering is done manually at the store-level. • Facility managers are time-constrained (and short on manpower). • Skills required for effective inventory control and forecasting are poor.
Extent to which underage and overage costs are recognized, measured, and internalized	• Inventory control and accounting systems enable precise estimates of overage/underage costs. • Management cost control system create accountability around overage costs. • Service level requirements create accountability around underage costs. • Buyback, consignment inventory contracts decrease the overage costs (Hackett, 1993)	• Inadequate inventory control and accounting systems lead to overage/underage costs not being recorded and used to support decision-making.
Store transshipment	• Transshipment among competitors is unheard of. It is difficult to manage, even within stores that belong to the same chain.	• Stores routinely borrow from neighboring stores, even when they might theoretically be considered competitors.

Table 4: Differences between mainstream retail inventory management assumptions and inventory management realities in private healthcare facilities in LMICs.

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7 Conclusions and Further Research

Our results shed light on differences between inventory management in mainstream retail supply chain management models and medicine supply chain management in settings where resources are limited. Specifically, our results also add nuance to several prevalent beliefs about medicine supply chains in LMICs. Being aware of these nuances can be crucial in streamlining the efforts being made to achieve universal health coverage. The first belief our analysis challenges is that poor availability at the facility level is caused mainly by supply-related issues (see, for example, Muiruri and Mugambi, 2017, and our literature review). Our results show that facilities in urban settings have access to an abundance of suppliers, and that supply-related issues are not the cause of low availability or understocking in private healthcare facilities (see Prop. 1).

The second is the rather widespread belief that IT provides a solution for medicine supply chain challenges in LMICs. Though IT definitely has great potential there (see Matchaba, 2019; USAID, 2019; USAID— Deliver Project, 2011), our results show that this applies only to a limited extent. Several facilities in our sample had invested in inventory control software, but had not observed the potential benefits.Our findings show that IT systems are often not fit-for-purpose or not consistently used, and many system functionalities are not being fully utilized. The basic skills and processes required for successful implementation are often lacking, and so too is buy-in from staff.

Facilities and organizations working towards UNSDG #3 and trying to improve medicine inventory control through IT need to realize the importance of tackling these issues. In line with Yadav (2015), we observe that the return on investment in inventory control systems is heavily influenced by factors such as such as number of products a facility holds, its weekly revenue, and training provided to the staff. There is a need to study this in much greater detail in order to make evidence-based decisions on investing in inventory systems.

A third (latent) belief we challenge is that overstocking is not an important problem, as suggested by the fact that our interviewees hardly gave it any thought, and it has also largely been ignored in the academic literature. First, we estimate that overstocking incurs substantial holding costs (cf. Yadav et al., 2012). Overstocking for one product, for example, may tie up the capital required to avoid stock-outs of other products. Second, many facilities (72%; 28 out of 39) admit that they give alternative or substitute drugs whenever they do not have the

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prescribed medicine in stock. Disquietingly, a few facilities admit to changing their prescription behavior based on what drugs they have in stock. This leads us to hypothesize that overstocking incentivizes facilities to engage in *demand shaping*. Such practices can clearly be harmful for patients and should be investigated through further studies. If they are confirmed by follow-up studies, appropriate laws and policies may be required to deter such behavior.

A fourth belief is that low *availability* is harmful for the facility, as is typically assumed in inventory management theory (Axsäter, 2015). Simulations by Yadav et al. (2012) confirm this in a context in which travel times to suppliers are long (several hours). However, we find that when a facility runs out of stock of a certain drug, it does not necessarily lead to lost sales. The facility can often backlog demand (due to short lead-times), substitute another drug, or get supplies from neighboring facilities (see Proposition 2).

Conventional inventory management wisdom suggests that facilities' ordering decisions depend on supplier stock levels. The short lead times reported by facilities in our study are an indication that suppliers are holding sufficient stock. However, we know that the preference for centralized holding of inventory at the suppliers vs decentralized holding at healthcare facility is determined by market search costs (Anupindi and Bassok, 1999). Given our finding that healthcare facilities share stocks with each other and can backlog demand to some degree, it warrants further examination of optimal inventory levels for facilities in this setting.

Finally, we question the rather widespread belief that low availability at the level of the individual retail healthcare facility has major health implications for patients (Elbireer et al., 2011; Pasquet et al., 2010; Mendis et al., 2007). We believe that in settings where clients have access to many facilities without significant transportation or physical barriers, where facilities themselves have access to many suppliers, and where supply lead-times are short, the impact of stock-outs at an individual retail facility may be less severe. More studies are needed to determine at which level of aggregation stock-outs lead to poorer health outcomes, but the issue is extremely pertinent, given the substantial resources invested in addressing these perceived challenges.

Our findings also raise questions about the prevailing practice of quantifying availability through stock levels at healthcare facilities, as recommended by the World Health Organization (2008). Holding in stock only 20% of items on the Kenya Essential Medicines List may sound

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rather problematic, but may actually be adequate in an urban context like Nairobi. We strongly believe that availability studies should also take into consideration lead-times and informal inventory-pooling networks. Availability at the supplier level or the percentage of client demand met within a specified time frame may be more informative metrics.

Our analysis also suggests that market characteristics partly explain both understocking and overstocking. They also seem to have a substantial negative impact on budget and consequently on service quality. If this is indeed the case, this is a strong argument for rationalizing catchment areas, as also advocated by Abuga et al. (2019). However, competition could also make facilities "listen to their clients" better and may clearly improve the quality of their service. Whether and when such benefits outweigh the disadvantages remains an important question, and future research on this is needed.

Many organizations aim to achieve UNSDG #3 and related goals by addressing the managerial causes and resource limitations revealed by our analyses: budget, skills, and systems. Future research should try to identify the best ways to mitigate the impact of credit constraints on ordering. Organizations such as the Medical Credit Fund provide small and medium-sized healthcare providers with fast mobile loans to deal with cash problems or to use for investments (UNSDG #3.8 and #3.C). Groups like mPharma and Shelf Life also provide inventory to the facilities on consignment. The loans alleviate credit constraints while allowing the retailer determine what quantities to order, whereas the consignment facility gives the suppliers more control over order quantities. Which approach is best is both a modeling question and also a topic for a future empirical study.

Dozens of pharmaceutical management systems exist, including open source ones, and many organizations offer inventory management training (UNSDG #3.C). We suggest that developing more programs should not be the priority at present. It is more important to investigate what prevents facilities from utilizing their existing systems effectively, as observed in our study. Our workshop, for example, revealed that staff turnover is a key barrier to investing in training of employees. Training was also described as being "too scientific." Another barrier is that facilities have not seen evidence of a positive return from their investment in IT systems and training. Future research presenting a holistic analysis of such issues is very much needed.

A related open question is whether there is a natural hierarchy of solutions to the six types

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of issues we have analyzed. For instance, is it the case that finding a solution to one issue will have little effect if other issues have not been addressed? Should one resolve budget, time, and motivation constraints before working on skills and systems? Should one would work on developing skills before enhancing systems? Which improvements should be made in parallel as opposed to sequentially? One hypothesis that follows from our observations, for example, is that the impact of system improvement may depend strongly on skills and motivation. Similar to existing research on improvements in manufacturing performance (see, for example, Ferdows and De Meyer, 1990; Noble, 1995, and references within), more empirical research assessing the link between causes and effective use of inventory control would help in determining the optimal sequence of improvement efforts.

We acknowledge that, despite our best efforts, there are certain limitations to our study. The descriptive nature of our study allowed us to account for situational complexities and provide deep and rich theoretical descriptions of the phenomena (Gioia et al., 2013). However, the number of observations is too small to study any heterogeneity of the phenomenon across sub-samples of private healthcare facilities. Furthermore, qualitative research is often labelled inappropriate to draw generalizable conclusions on (sub)populations: "the intent is not to generalize to a population, but to develop an in-depth exploration of a central phenomenon" (Creswell, 2009). This limitation is also reflected in our sample selection process (see Section 4), which was performed "purposively" to achieve information saturation, rather than "randomly" to achieve generalizability. Exploratory studies such as ours lead to hypotheses and propositions, and thereby lay the groundwork for future research. Future empirical research collecting data from a wider range of healthcare facilities and analyzing validity of our results for sub-samples is highly needed. We note that this phased approach to research – first perform an exploratory study, then perform confirmatory research – is common and advocated for studying new phenomena (Tukey, 1980).

Another limitation of our study is that all the interviews were conducted in Nairobi. Based on the similarity in context and our understanding of the on-the-ground realities, we believe that our insights are generalizable to other resource-constrained settings in LMICs. However, this still needs to be tested rigorously. In particular, field experiments and empirical analyses on data from different LMICs should be conducted to rigorously test the causal relationships

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we proposed in Section 6.

We hope that future researchers will build on the propositions that resulted from this analysis. A finer root-cause analysis of the mismatch between demand and supply in private healthcare facilities, along with an examination of its effect on patients and facilities, would throw much needed light on to this often overlooked link. Strengthening the private sector will improve the overall healthcare supply chain and help in making progress towards universal health coverage and, more generally, the UN's other sustainable development goals.

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A Search terms

Database	Search string
Pubmed	((TS=(demand supply mismatch OR overstocking OR understocking OR availability OR inventory* OR shortage OR oversupply OR over-supply)) AND (TS= (pharmac* OR health care facilit* OR healthcare facilit* OR healtch-care facilit* OR clinic* OR maternity home* OR nursing home*)) AND (TS = (medicine* OR drug*)) AND (TS=(Africa OR Emerging countries)) AND (TS= ("Bottom of the pyramid" OR poor OR low income OR low-income OR LMIC)))
Web of Science	(demand supply mismatch [Title/Abstract] OR overstocking [Title/Abstract] OR understocking [Title/Abstract] OR availability [Title/Abstract] OR inventory* OR shortage [Title/Abstract] OR oversupply [Title/Abstract] OR over-supply [Title/Abstract]) AND(pharmac* [Title/Abstract] OR healthcare facility* [Title/Abstract] OR health care facility* [Title/Abstract] OR clinic* [Title/Abstract] OR maternity home* [Title/Abstract] OR nursing home* [Title/Abstract]) AND(medicine* [Title/Abstract] OR drug* [Title/Abstract]) AND(Africa [Title/Abstract] OR Emerging countries [Title/Abstract] OR LMIC [Title/Abstract]) AND("Bottom of the Pyramid" [Title/Abstract] OR poor [Title/Abstract] OR low income [Title/Abstract] OR low-income [Title/Abstract])

B Pharmaceutical Terminology

The terminology used for facilities and personnel are borrowed from the Republic of Kenya's

- the Nurses Act, 1983
- The Medical Practitioners and Dentists (Inspections and Licensing) Rules, 2014
- The Clinical Officers (Training, Registration and Licensing) Act, 2017.393

For the reader's convenience, we redefine them below:

Facility definitions

- "Clinic" means a consulting office or room or an outpatient department without beds used by a medical or dental practitioner for the diagnosis and treatment of diseases or the giving of medical or dental advice and instructions.

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- "Hospital" means an institution which has, in addition to resident medical practitioners or dentists, an operating theater and a mortuary.
- "Maternity home" means any premises used for the reception and management of expectant women or of women who have given birth within the preceding six weeks.
- "Medical center" means any such approved health facility operated by a clinical officer with a bed capacity of not more than fifteen to provide preventive, promotive, curative and rehabilitative services.

Personnel definitions

- "Clinical officer" means a person who having successfully undergone a prescribed course of training in an approved institution is a holder of a diploma or degree certificate in clinical medicine and community health.
- "Registered nurse" means a person whose name appears in the register kept by the Registrar under section 12(1)(a).

C 2016 Questionnaire

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1 INTRODUCTION

The field agents will use this questionnaire to collect data from the contracted healthcare providers. The questionnaire will start with filling in a table with all relevant details of the drugs ordered that week. The agents will then ask a set of similar questions each week, which relates to the information filled in in the table. There is also a set of questions that will only be asked to the healthcare providers in weeks 1, 3, 6 and 8 of the data collection process. Most questions will only have closed-end answer possibilities in order to increase homogeneity of data and to prevent confusion with the interviewed healthcare providers.

1.1 Ordering details table

1.1.1 Healthcare provider input

The 50 contracted healthcare providers are requested to provide input on each ordered drug order per week in a table format:

i. Date
ii. Drug Molecule name
iii. Drug name (name of the pack)
iv. Supplier/wholesaler name
v. Drug manufacturer name
vi. Mono or combi therapy
vii. Dose form
viii. Strength
ix. Pack size
x. Number of packs ordered
xi. Number of packs received
xii. Price per pack in Ksh
xiii. Discount received
xiv. Items experienced stock out
xv. Items in stock
xvi. Retails sales price of drug in Ksh
xvii. KEML number
xviii. PPB registration number
xix. Brand or generic

A. Healthcare provider to specify the molecule/ active pharmaceutical ingredient for each ordered drug. There will be room for filling in multiple molecules per drug as certain drugs contain multiple active pharmaceutical ingredients (no need to fill in support ingredients, e.g. acid or starch)
B. Healthcare provider to mention the name of each ordered drug that is indicated on the pack
C. Healthcare provider to mention the name of the supplier/ wholesaler where each ordered drug has been procured from

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D. Healthcare provider to mention the name of the manufacturer of each ordered drug, in case this is different from the supplier/ wholesaler in question C
E. Healthcare provider to indicate whether the drug is a mono or a combination therapy (mono or combination are the answer options)
F. Healthcare provider to select from the following answers and fill in the table accordingly (based on the NFC coding system):
    1. Oral solid
    2. Oral liquid
    3. Parenteral ordinary
    4. Rectal systemic
    5. Other systemic
    6. Oral topical
    7. Topic dermal haemor external
    8. Ophthalmic
    9. Otic
    10. Nasal topical
    11. Lung administration
    12. Vaginal
    13. Non-human use and others
G. Healthcare provider to fill in the dose strength of the drug. For injectable please indicate mg/ml
H. Healthcare provider to fill in the pack size
I. Healthcare provider to mention the number of packs of this drug ordered
J. Healthcare provider to mention the number of packs of this drug received, in case this is different from what had been ordered
K. Healthcare provider to mention the price per ordered drug pack in Ksh, prior to:
    1. Any potential discount that has been received
    2. Any payment for delivery of drugs that has been made
L. If the healthcare provider has received a price discount, then the healthcare provider should mention the discount in % or as the number of drug items received as extra or bonus
M. Healthcare provider to fill in whether a stock out of this drug at the healthcare provider level was experienced prior to this order (yes or no are the answer options)
N. Healthcare provider to fill in the number of packs of this particular drug that had been in stock before this order
O. Healthcare provider to fill in the retail sales price per pack of this particular drug to the patient in Ksh

Field agents are requested to take photocopies of receipts with their tablets in order for the pharmacist to cross check input given by the healthcare provider.

1.1.2 Pharmacist Consultant input

P. Pharmacist consultant to reference each ordered drug to the specified unique drugs code, according to the Kenya Essential Medicines List published by the Ministry of Medical Services and Ministry of Public Health & Sanitation.

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Q. Pharmacist consultant to input Pharmacy & Poisons Board ("PPB") registration number of each unique drugs supplier mentioned by the healthcare provider.

R. Pharmacist consultant to indicate whether the drug item is an original brand, a branded generic or a generic drug type

1.2 Additional questions around the order and procurement process

1.2.1 Questions to be answered in weeks 1, 3, 6 and 8 of the data collection phase

1. **Who is responsible for drugs ordering at your facility?**
    a. Choice answers:
        i. Pharmacist
        ii. Pharmaceutical technologist
        iii. Pharmacy assistant
        iv. Doctor
        v. Nurse
        vi. Other – healthcare provider to indicate role

2. **Under which license are the drugs procured in your facility?**
        i. Pharmacist
        ii. Pharmaceutical technologist
        iii. Pharmacy assistant
        iv. Doctor
        v. Nurse
        vi. Other

3. **How does the order process typically work at your facility?**
    a. Healthcare provider to allocate from a total of 100 points to these attributes:
        i. Healthcare provider phones supplier
        ii. Supplier phones healthcare provider on a recurring basis
        iii. Healthcare provider visits supplier(s)
        iv. A sales representative visits my facility
        v. Healthcare provider emails supplier
        vi. Supplier emails healthcare provider on a recurring basis
        vii. Via facility IT system that automatically generates an order
        viii. Other – healthcare provider to specify answer

4. **How often do you normally procure drugs?**
    a. Choice answers:
        i. Daily
        ii. Every few days
        iii. Weekly
        iv. One larger order per month, with top-ups weekly
        v. Differs, as required

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5. **How do you mainly decide what brand to procure for a particular drug molecule?**
    a. Healthcare provider to allocate from a total of 100 points to these attributes:
        i. Price
        ii. Brand
        iii. Quality
        iv. Efficacy
        v. Availability
        vi. Patient preference

6. **What are the attributes by which your patients purchase their drugs from you?**
    a. Healthcare provider to allocate from a total of 100 points to these attributes:
        i. Price
        ii. Brand
        iii. Quality
        iv. Efficacy
        v. Availability
        vi. Patient preference

7. **How do you determine to buy original branded vs branded generics vs generic drugs?**
    a. Healthcare provider to give an answer to the open question

8. **What is your main motivation to procure from your current supplier(s)?**
    a. Healthcare provider to name their supplier(s) and answer for each supplier mentioned
    b. Healthcare provider to allocate from a total of 100 points to these attributes:
        i. Price
        ii. Quality
        iii. Variety
        iv. Relationship
        v. Provision of credit
        vi. Speed of delivery
        vii. Reliability of supply
        viii. Accessibility
        ix. Other – healthcare provider to specify answer

9. **How likely are you to recommend this drug supplier to a colleague?**
    a. Healthcare provider to name their supplier(s) and answer for each supplier mentioned
    b. Healthcare provider to give a score between 1 (not at all) to 10 (definitely recommended)
    c. Healthcare provider to indicate a reason for its answer at b

10. **Do you think your current supplier(s) provide quality and how have you determined this?**
    a. Healthcare provider to name their supplier(s) and answer for each supplier mentioned
    b. Choice answers:
        i. I think my supplier is of high quality given the company is properly licensed
        ii. I have ordered from this supplier before and my patients have never been disappointed with the quality, so it must be good

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iii. I do not know exactly what the quality of the drugs that I procure is
iv. Sometimes I doubt the quality of this supplier, as I have experienced receiving similar drugs at different orders that are of different quality

11. What would you look for in a potential new drugs supplier?

a. Healthcare provider to allocate from a total of 100 points to these attributes:
   i. Price
   ii. Quality
   iii. Variety
   iv. Relationship
   v. Provision of credit
   vi. Speed of delivery
   vii. Reliability of supply
   viii. Accessibility

12. Do you have a legal contract in place with your current supplier(s)?

a. Healthcare provider to name their supplier(s) and answer for each supplier mentioned
b. Choice answers:
   i. Yes
   ii. No

13. How was the relationship with your supplier(s) established?

a. Healthcare provider to name their supplier(s) and answer for each supplier mentioned
b. Choice answers:
   i. I had heard of them and thus reached out to this supplier myself
   ii. The supplier called me
   iii. A sales representative of the supplier visited my facility
   iv. Other facilities in the neighborhood are using this supplier, so it must be good
   v. Other – healthcare provider to indicate answer

14. How do you maintain drugs supply chain information (e.g. order details, number of packs in stock)?

a. Choice answers:
   i. In a file
   ii. In a notebook (written)
   iii. IT system (computer software)
   iv. IT system (computer word or excel file)
   v. I do not currently maintain this information
   vi. Other – healthcare provider to indicate answer
b. If answer ii is selected, please indicate which software program is used

15. How do you determine which items to keep in stock?

a. Healthcare provider to allocate from a total of 100 points to these attributes:
   i. Popular demand from my patients
   ii. Disease trends/seasonality of diseases

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iii. I keep a general minimum stock level of certain drugs based on the Kenya Essential Medicines List
iv. Availability of the stock at the supplier
v. Lack of past availability and therefore need to stockpile
vi. Other – healthcare provider to indicate answer

16. **Do you keep track of expiry dates of the drugs you order?**
    a. Choice answers:
        i. Yes – tracked in a notebook (written)
        ii. Yes – tracked with an IT system (computer)
        iii. Yes – tracked in another way – healthcare provider to indicate the way
        iv. No – I do not track this pro-actively

17. **How do you deal with expiry of drugs that you ordered?**
    a. Healthcare provider to allocate from a total of 100 points to these attributes:
        i. I return the expired drugs to the supplier where I bought it from
        ii. I have to throw these drugs away
        iii. Whenever a drugs item is about to expire and I cannot sell it to my patients I am able to trade it with other local facilities
        iv. In general, I do not experience expiry of drugs that I have ordered
        v. Other methods used – healthcare provider to indicate method

1.2.2 Weekly questions

1. **What payment method did you use to procure drugs items from your supplier(s) this week?**
    a. Healthcare provider to name their supplier(s) and answer for each supplier mentioned
    b. Choice answers:
        i. Bank payment
        ii. Cash payment
        iii. M-Pesa payment
        iv. Combination of the above – healthcare provider to indicate split of methods used in (allocating to 100 points)

2. **Did you receive credit when procuring drugs this week?**
    a. Healthcare provider to name their supplier(s) and answer for each supplier mentioned
    b. Choice answers:
        i. Yes – Healthcare provider to indicate credit terms by indicating the number of days (e.g. 30 or 60 days of credit)
        ii. No

3. **How quickly did you pay your supplier(s) this week from the point at which they lent you stock on credit? And what was your motivation for this?**
    a. Healthcare provider to name their supplier(s) and answer for each supplier mentioned
    b. Choice answers:
        i. I paid upfront

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ii. I paid upon receiving of the drugs
iii. I pay according to the credit terms (e.g. 30 or 60 days see question 2), making sure I stay within the number of credit days offered
iv. I pay right before making a new order with this supplier
v. Other – healthcare provider to specify answer

c. Healthcare provider to indicate reason for answer in b.

4. What delivery method has been experienced this week?

a. Healthcare provider to name their supplier(s) and answer for each supplier mentioned
b. Choice answers:
   i. Delivery to healthcare provider's facility
   ii. Pick-up at supplier
   iii. Pick-up at central warehouse

5. What means of transport was used to deliver the drugs?

a. Healthcare provider to name their supplier(s) and answer for each supplier mentioned
b. Choice answers:
   i. Van from the supplier
   ii. Car
   iii. Motor bike
   iv. Public transport
   v. Courier transport
   vi. Bicycle
   vii. On foot (walking)
   viii. Other

6. Did you procure any temperature sensitive drugs this week? If yes, where do you store them in your facility? And lastly, how were these drugs delivered to you/ picked up by you?

a. Healthcare provider to answer with yes or no
b. If the answer is yes at a, the healthcare provider should indicate where these drugs are held at the facility
c. If the answer is yes at a, the healthcare provider should indicate how the drugs were delivered/ picked up (in a cooled box or other means for example)

7. What does the drugs supplier charge you for delivery of drugs?

a. Healthcare provider to name their supplier(s) and answer for each supplier mentioned
b. Choice answers:
   i. I am charged separately – healthcare provider to indicate price in Ksh
   ii. I am charged but I believe this is calculated in the drugs prices
   iii. I believe I am not charged

8. Did you receive all drugs you ordered this week?

a. Healthcare provider to answer yes or no
b. If answer is no at a, the healthcare provider could specify a reason for the difference in what was ordered and what actually received

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9. **What delivery lead times were experienced this week?**
    a. Healthcare provider to name their supplier(s) and answer for each supplier mentioned
    b. Choice answers:
        i. Same day delivery when order placed early in the morning
        ii. Next day delivery
        iii. More than 2 days
        iv. More than 5 days or more

10. **Were lot/batch numbers of drugs items visible on order or delivery receipts of this week?**
    a. Choice answers:
        i. Yes they are visible on all drugs items I have ordered
        ii. Yes on most drugs items I have ordered they are visible
        iii. On some drugs items I have ordered they are visible
        iv. No, they are not visible on the drugs items I have ordered

11. **In the last week, did you experience stock-outs and if yes what could be the reasons for this?**
    a. Healthcare provider to answer with yes or no. If yes, please specify the reason using the answer options at b
    b. Healthcare provider to allocate from a total of 100 points to these attributes:
        i. High consumption of drugs by my patients
        ii. Stock-outs from suppliers
        iii. I ordered the drugs item too late
        iv. The supplier has delivered the drugs item too late
        v. I do not have enough storage space for certain drugs items
        vi. Insufficient money to purchase products
        vii. Some drugs items are too expensive to keep in stock
        viii. Short shelf life of certain drugs items
        ix. In general, I do not experience stock-outs
        x. Other reason, healthcare provider to specify the reason

12. **By what means have your patients paid mostly for a drug item in the past week?**
    a. Healthcare provider to allocate from a total of 100 points to these attributes:
        i. Cash
        ii. M-Pesa
        iii. Bank transfer
    b. If answer II is selected, healthcare provider is to indicate whether a payment is made to an M-Pesa number, till number or paybill number. Please also indicate the paybill number if that is used

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D Formulating the interview guide

Our survey questionnaire was put together following a commonly accepted methodology, formalized by Kallio et al. (2016). This methodology consists of five phases:

1. Validating the appropriateness of semi-structured format
2. A thorough investigation and retrieval of previous knowledge
3. Formulating a preliminary semi-structured interview guide
4. Pilot testing the preliminary guide
5. Presenting the complete semi-structured interview guide

In our review of extant literature (see Section 3), we discuss in detail about the lack of prior research on private healthcare facilities in LMICs. The primary objective of this research is to fill that gap by asking, "*Why is there a mismatch between demand and supply?*", and formulating new hypotheses that lay the groundwork for future research. In short, our research is exploratory and the exact nature of the theory that would shape up from this study was not known beforehand. When asking such exploratory questions, taking the context into consideration is crucial, and qualitative studies such as ours are seen as most appropriate (Baxter et al., 2008).

We chose a semi-structured interview format for our study because it are considered to be "flexible, accessible, intelligible and capable of disclosing important and often hidden facets of human and organizational behavior" (Qu and Dumay, 2011). It is often considered to be the most effective and convenient means of gathering information (Kvale and Brinkmann, 2009). Most importantly, semi-structured interviews enable interviewees to provide responses in their own terms and in the way that they think and use language. These qualities make semi-structured interviews the ideal format of data-collection for our exploratory study.

Having validated semi-structured interviews as the appropriate format for our objectives (step one), we retrieved relevant previous knowledge (step two) through a a systematic and thorough review of literature relating to over- and understocking of medicines in LMICs. We searched the PubMed and Web of Science databases for articles written in English within the last 20 years (going back from 1 October 2018), and available in full-text. Search terms (see

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Appendix A) were chosen so as to find papers relating to 1) medicines, 2) over-/understocking, 3) healthcare facilities, and 4) low-income settings. Two authors independently assessed the abstracts of the resulting 150 articles.

Based on causes suggested in the literature,we formulated a preliminary semistructured interview guide (step three). It was specifically designed to assess each facility's 1) inventory control systems, 2) inventory control skills, 3) time/human resource constraints, 4) budget constraints, 5) motivations for inventory control, and 6) relationship/issues with suppliers. Several open questions were included to capture potential causes not covered by these six categories. The interview guide was reviewed by each author and two other academics with advanced knowledge on the topic. We then proceeded to pilot test the guide (step four) during 3 facility visits in August 2018 and finalized it afterwards (e.g., we aligned the jargon with that of the interviewees). We present the final version of the interview guide (step five) in Appendix E

E 2018 Questionnaire

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Draft Questionnaire Behavioral/Managerial Aspects of Drug Inventory Control

PharmAccess & University of Nairobi &
INSEAD Humanitarian Research Group

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Objectives

1. To analyze root causes for overstocking and understocking of drugs in Nairobian privately owned health facilities.
2. To understand how inventory managers in privately owned health facilities make inventory management decisions, how they deal with inventory management policies and inventory management systems.

Format

Semi-structured interview with inventory managers. I.e., the interviewer uses a baseline list of themes and questions but has the freedom to deviate from that to discuss relevant issues that arise during the interview/ to skip themes/questions that are not (fully) relevant.

Themes to be explored

1. Current inventory management practices
2. Root causes of over- and understocking and their relative importance
    a. Incentives
    b. Skills
    c. Time/priorities
    d. Supplier issues
3. Adherence to inventory control policies
4. Use of inventory control software
5. Social relationships between pharmacists – competitive vs. cooperative behavior

Question list

N.B. the pieces described in italic describe what follow-up questions should focus on.

Theme 0: Introduction questions

1. **Characteristics of the interviewee – job title, roles, years of experience,…**
    a. **What is your highest complete degree?**
    b. **How many years have you been performing pharmaceutical activities?**
    c. **How many years have you worked in the health facility?**
    d. **Is your health facility part of a chain/network?**

2. **On a scale of 1 – 10, how would you rate the way drug inventories are currently being managed? Please explain your answer.**
   Ask the interviewee for a qualitative justification of the numerical score they give. Follow up with any challenges they identify in their justification.

3. **Do you think you have too much stock of certain medicines?**

4. **Do you think you have too little stock of certain medicines?**
   These are fundamental questions designed to ask the respondant if they see a systematic mismatch between demand and supply.

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Theme 1: Current inventory management practices

5. **How do you manage your inventories?**
   Identify through follow-up questions what processes and decision rules are in place:
   - *Policy/ intuition-based*
   - *Periodic review/ continuous review.*
   - *Min-max/ forced-ordering/ two bin/ order for customer*
   - *Stock cards/ IT system*

6. **How many people in your facility are involved in inventory control? Who is responsible for what?**
   Who makes the ordering decisions? Who places the order? Who takes care of shelving the received goods?

7. **[If policies in place] How do you set your inventory control rules?**
   Identify whether a systematic calculation method is being used to select which items to keep on stock and to set reorder points and order quantities. If so, identify what this method is and how it was obtained (training/ IT system/ …).

8. **[If policies in place] When do you update your inventory control rules?**
   Identify whether there is a regular updating and whether there are specific triggers for that (stockouts/ stock obsolescence).

9. **[If intuition-based] How do determine how much to order and when to order?**
   Identify the heuristic rules/guidelines being applied.

10. **What inventory-related data do you routinely collect?**
    Check for stock keeping, transaction, expiry date, consumption records, and notes on adverse effects of medicines. Try to take pictures/describe the details.

11. **Can we have a look at the notebook where you keep an overview of your inventory – orders, stocks, expiration dates?**
    Try to take pictures or describe the details of what you see in the notebook – how comprehensive is it? How well maintained is it? Any obvious mistakes in judgement.

12. **If you encounter a situation where you have too much or too little stock of a given drug, do you put efforts in identifying the root causes?**
    Identify how it affects future inventory control decisions.

Theme 2a: Root causes of over- and understocking – Incentives

13. **Why is it important for you to have drugs on stock?**
    Identify to what extent inventory control is driven by economic incentives vs. by patient incentives.

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14. **When determining which drugs to keep on stock; what are your major considerations? What is your forecasting policy? Do you have any guidelines that you follow?**
    Identify the importance of demand rates, holding costs, stockout costs, drug essentiality (KEML), budget, space limitations, promotions,…
    N.B., this question has been asked in first study as well, but it is probably worthwhile to go in a bit more depth. Same for next question.

15. **When determining what drug quantity to order; what are your major considerations?**
    Identify the importance of demand rates, holding costs, stockout costs, drug essentiality (KEML), budget, space limitations, promotions Reliability of suppliers? Gaming? Level of coordination with nearby facilities?

16. **What does a client typically do when a drug is out of stock?**
    Identify the implications of a stockout. Will the patient quickly manage to find the drug elsewhere? If so, at what type of facility (drug vendor/chemist, corner shop, hospital, etc.)? Or will the patient return to the pharmacy? Will it affect the facility's goodwill? What does it cost patient in terms of health, time, inconvenience, ..?

17. **Are there any drugs that you always try to maintain "in-stock"? How do you choose which drugs are treated this way?**
    How do they identify which drugs are important to be always kept in stock? Do they follow the KEML list? Or do they have other concerns, such as the margins on the medicine or the demand patterns for that medicine?

18. **What does it cost you when drugs expire?**
    Identify whether they receive compensation when they return drugs to the supplier or loss due to selling it for a low price or throwing it away.

19. **Does your facility monitor the performance of inventory control?**
    Check whether any quantitative performance measures are in place (e.g. fill-rates/inventory turnover).

20. **What happens in case of good or bad performance?**
    Check for positive and negative incentives.

21. **Do you have stock losses in your pharmacy?**
    Ask for reasons for stock loss. Is it because of pilferage? Or other storage/stock-keeping related reasons?
    Ask for an estimate of the percentage of drugs stolen and identify whether this forms a barrier towards implementing transparent systems.

Theme 2b: Root causes of over- and understocking – Skills

22. **[If applicable] During the first study, you/your colleague mentioned that "High consumption of drugs by my patients" is a primary reason of stockouts. Could you elaborate on this?**

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Identify whether demand is really the issue or whether it is a matter of inadequate inventory control rules.

23. **How does the changing seasons/weather affect demand for drugs in your pharmacy? How do you plan for this? Do you take this into account when making inventory decisions?**

    The idea here is to understand how they deal with seasonality in demand. If they consciously account for it in making inventory decisions.

24. **How did you learn how to manage drug inventories?**
    Identify if any formal training was obtained.

25. **What would you like to learn about inventory management if you were invited to a training session?**
    Identify which aspects of inventory management are weak due to a lack of skills.

Theme 2c: Root causes of over- and understocking – Time/priorities

26. **Besides managing inventories, do you have any other responsibilities? If so, how do you allocate your time between them?**
    Identify the respective priority of inventory management vs. other responsibilities

27. **Do you think that inventory management would improve if you had more time for that?**
    Identify what goes wrong due to a lack of time.

Theme 2d: Root causes of over- and understocking – Supplier issues

28. **[If applicable] During the first study, you/your colleague mentioned that "Stock-outs from suppliers" is a primary reason of stockouts. Could you elaborate on this?**
    Identify how prominent the problem is, whether this is accounted for in inventory control rules, and whether multiple suppliers are available for the same product. Also identify what they do in case of a stockout and why they think suppliers are facing stockouts.

29. **Do suppliers sometimes push facilities to buy more drugs than they actually need? Does this cause any problems?**
    Identify how they push (e.g., promotions/ Christmas gifts/ other incentives) and whether this leads to overstocking and obsolescence.

Theme 3: Adherence to inventory control policies

This section is only relevant for facilities which indicate they have inventory control policies.

30. **You indicate that your facility has inventory control rules. How often are these rules strictly obeyed?**
    Ask for an estimate of the percentage of cases when the moment an order is placed is different from the one specified by the rule and the percentage of cases when the ordered quantity is different from the one specified by the rule.

31. **For what reasons do you sometimes place orders sooner/later than the rule tells you to do?**

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Identify root causes

32. **For what reasons do you sometimes order more/less than the rule tells you to do?**
    Identify root causes

Theme 4: Use of inventory control software

This section is only relevant for facilities which indicate to have inventory control software (either in the baseline study or now). Check beforehand the functionalities of the software.

33. **You indicate that your facility has installed inventory software. How do you use this software?**
    a. **Why did you start using this software?**
    b. **How did you finance the software?**
    c. **How did you choose the software among the available options?**

    Identify whether it is used as a digital stock card, for determining order quantities and reorder points, placing the order, expiration dates/batch numbers.

34. **Is the software consistently used? If not; why?**
    Identify whether data entry is consistent and ask for root causes of inconsistencies. E.g., system complexity, transparency issues, technical breakdown, technical support,…

35. **Do you feel comfortable (well-skilled) in using this software?**
    Ask whether any formal training way obtained.

36. **[If applicable] Why do you presently not use the software to keep track of inventory levels?**
    Identify root causes

37. **[If applicable] Why do you presently not use the software to determine suitable order quantities and reorder points?**
    Identify root causes

38. **[If applicable] Why do you presently not use the software to place orders?**
    Identify root causes

39. **[If applicable] Why do you presently not use the software to keep track of expiration dates/batch numbers?**
    Identify root causes

Theme 5: Social Structure

40. **How often do you get a chance to interact with other pharmacists in your region? What are the circumstances in which these interactions typically happen?**
    Sometimes conferences and meetings are organized by suppliers or professional societies/associations such as PSK, KMA, and NNKA. Do they actively try to maintain

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relationships with other pharmacists in the region? What steps do they take to maintain these relationships?

41. **Do you think there are too many pharmacies in your neighbourhood?**
    This question is supposed to elicit a response about how they perceive the other pharmacists in their region? Do they look at them as competition or do they consider them to be part of a community that functions cooperatively?

Theme 6: Closing questions

42. **Are there any inventory management themes in the pharmacy business that we missed in this questionnaire?**
    Open ended question to improve the questionnaire and edit it when speaking to subsequent pharmacists.

43. **If you had more capital for your business, what would you do?**
    This question has to be asked in a format that elicits a response without bias about interventions/investments the pharmacies perceive as most cost-effective. Alternatives:
    Where do you see the opportunity for improvement in your business?
    What would improve your pharmacy business?

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